

JOSEPH A. WALTON

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RESEARCH INTERESTS

- Operations Research; reinforcement learning for queueing control; stochastic optimization; simulation; semi-Markov decision processes; machine-interference and repair-control systems.

EDUCATION

- **Missouri University of Science and Technology** - Ph.D. student, Engineering Management and Systems Engineering Kummer Innovation & Entrepreneurship Doctoral Fellow; GPA: 4.0 (reported after first year).

SELECTED RESEARCH

- **Empirical Evaluation of Policy-Gradient Reinforcement Learning for Dynamic Service Control in an M/M/1 Queue**
Conference paper presented at the IISE Annual Conference & Expo, 2026. Coauthor: Gabriel Nicolosi.

- **State-Action Space Growth in Discrete Machine-Interference Repair Control with Speed-Dependent Failures**
Working paper in preparation. Examines exact dynamic-programming boundaries, state representation, aggregation risk, and scalable ADP/RL approaches.

TEACHING AND WORKSHOPS

- Led two 2.5-hour solver workshops covering Julia/JuMP/HiGHS and Python/PuLP, including formulation, implementation, solution interpretation, and debugging.
- Selected to support Ph.D. seminar activities within the department.

TECHNICAL METHODS

- Python (SimPy, Gymnasium, PyTorch, PuLP); Julia (JuMP, HiGHS); R (tidyverse); discrete-event simulation; MDP/SMDP modeling; relative value iteration; policy-gradient reinforcement learning; queueing theory; mixed-integer optimization.

PRIOR PROFESSIONAL EXPERIENCE

- Process Improvement Manager in glass manufacturing, with experience connecting operational problems, data, process design, and decision support.

SELECTED PRESENTATIONS

- IISE Annual Conference & Expo, 2026 - Oral presentation on policy-gradient reinforcement learning for dynamic service control.
- Research posters and departmental presentations on reinforcement learning, simulation, and stochastic control.